

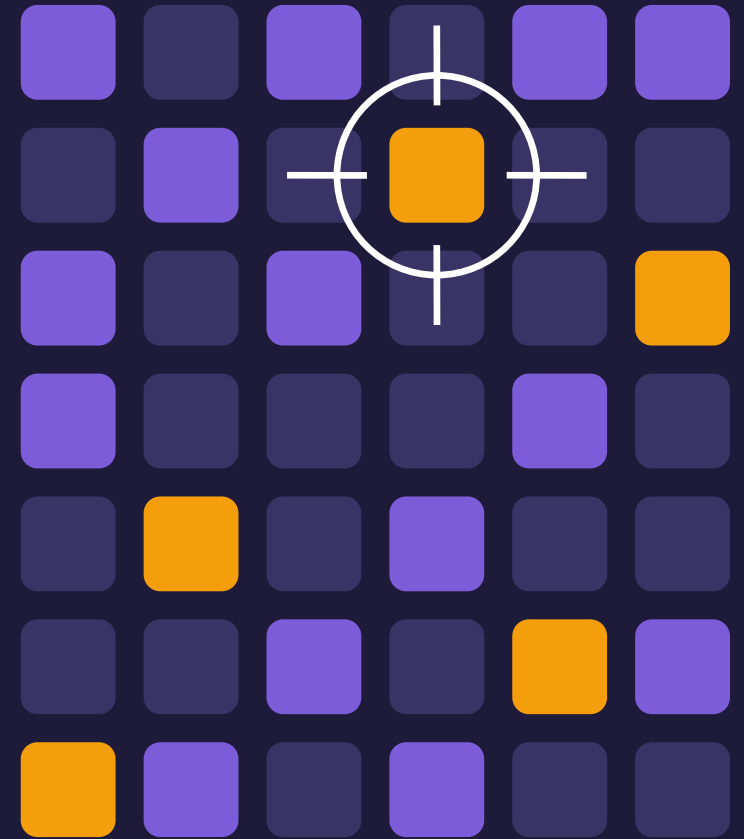
IBM BOB 2.0 HACKATHON

MutantHunter

Finds the bugs a test suite misses, writes tests that catch them, and proves the gain on mutants the agents never saw.

An IBM Bob skill plus a deterministic Python harness, run on the python-validators library.

Evan tool, skill, test generation **chu** evaluation, dashboard, video



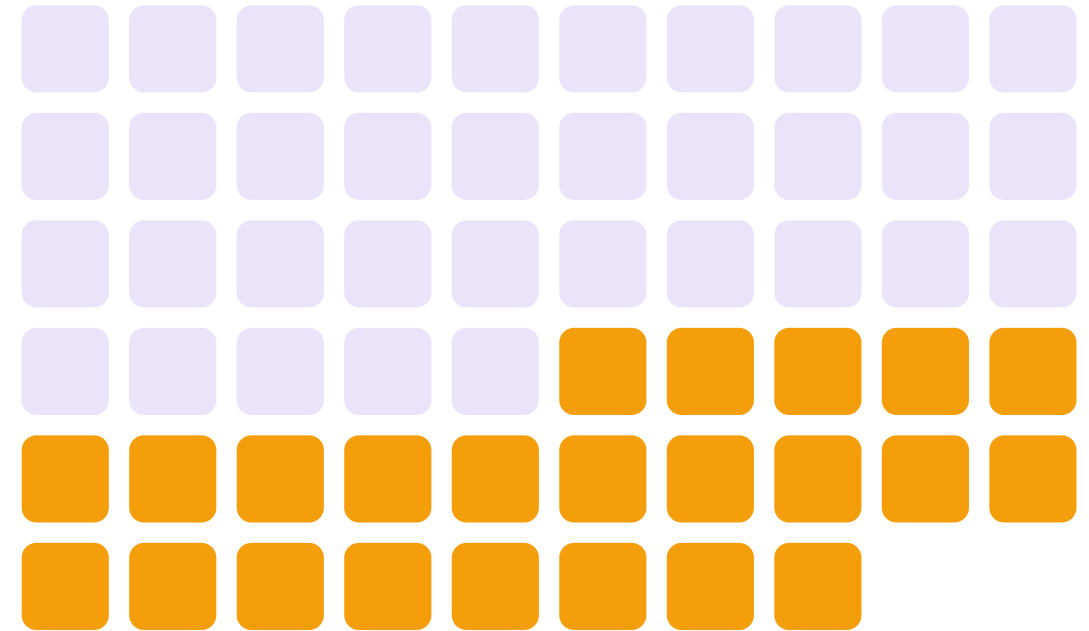
Coverage is not the same as catching bugs

91.9%

line + branch coverage of url.py

23 of 58

injected bugs (mutants) survive its human tests



35 killed

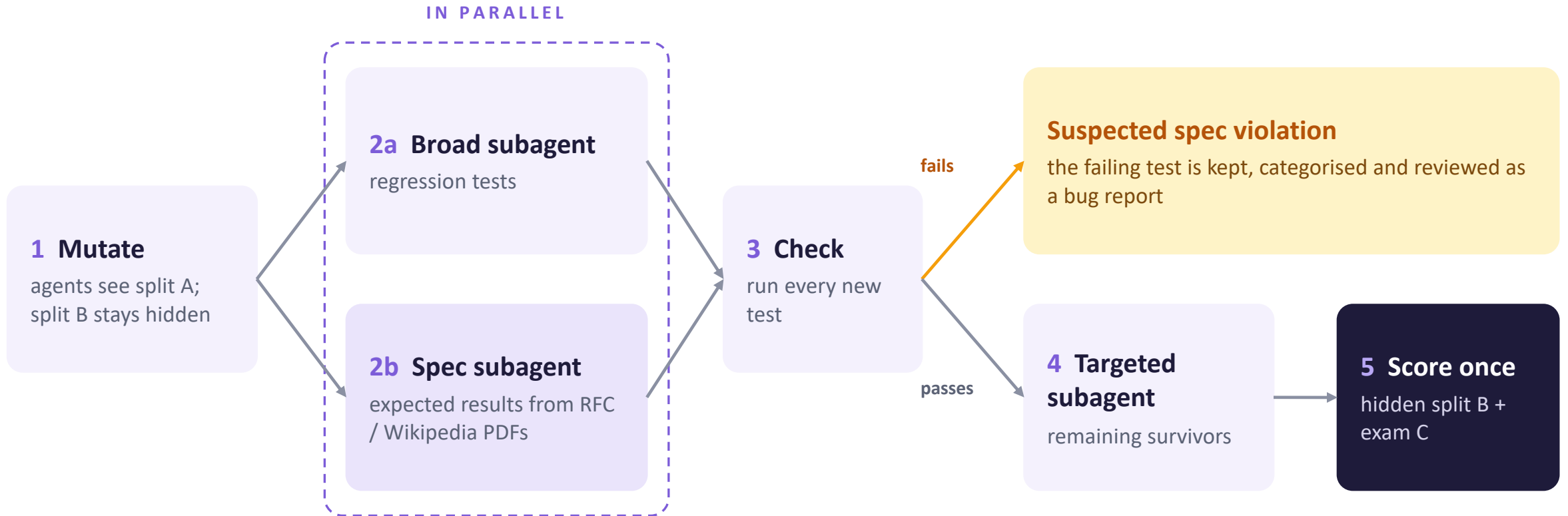
23 survived

url.py · 58 mutants



A plain “write more tests” prompt pins today’s behaviour, bugs included: in finance it wrote tests around the broken ISIN check digit.

One sentence in Bob runs the whole loop



› Use the `mutant-hunter skill` on the `module url`.

The harness is deterministic and never calls an LLM.

Six Bob features, each doing real work



1 task

Plan mode

wrote the build plan, sub-tasks ST1–ST9



Product code

Agent mode

harness, stats, dashboard, 128 unit tests



1 sentence

Skill

runs the whole pipeline on a module



1 rules file

Rules

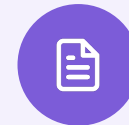
spec first; failing tests are never deleted



13

Parallel subagents

broad + spec side by side, targeted on url



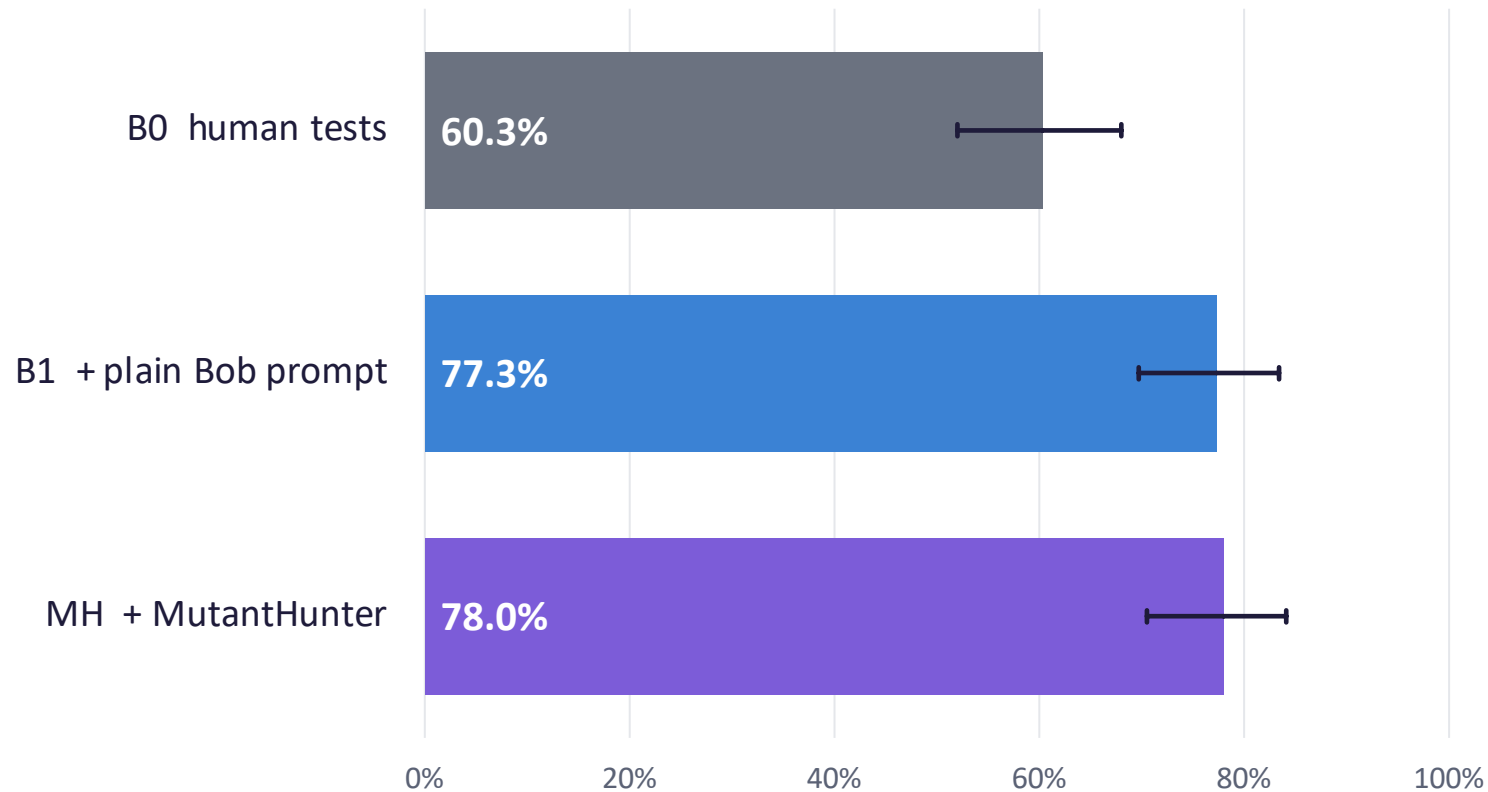
10 → 137

Document understanding

spec PDFs turned into numbered rules

22 Bob tasks (Evan 8, chu 14) · ≈88.4 Bobcoins · every session exported in bob_sessions/

Hidden split B: an honest tie with the plain prompt



Kill rate on 141 hidden mutants, 4 main modules. Whiskers: 95% Wilson CI (B0 52.0–68.0, B1 69.7–83.4, MH 70.5–84.1).

KILLED BY ONLY ONE SUITE

2

B1 only

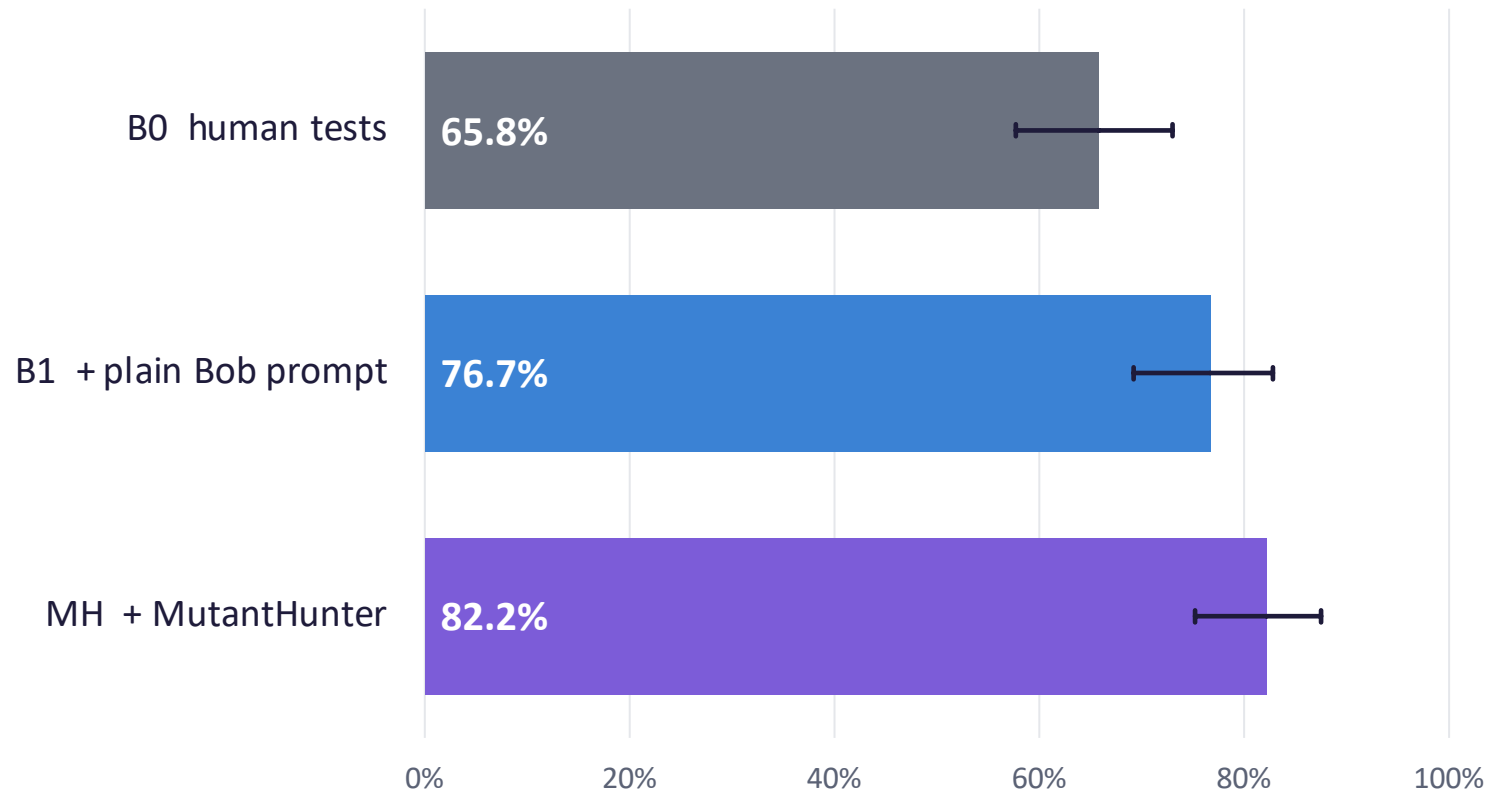
3

MH only

McNemar $p = 1.00$

Both AI suites add ~17 points over the human tests. On split B, MutantHunter does not beat the plain prompt.

Exam C, unseen operators: a trend toward MH



Kill rate on 146 mutants from operators that did not exist while the tests were written. Whiskers: 95% Wilson CI (B0 57.7–73.0, B1 69.2–82.8, MH 75.2–87.5).

KILLED BY ONLY ONE SUITE

1

B1 only

9

MH only

McNemar $p = 0.0215$

Not significant after Bonferroni correction over 4 tests, so a trend: spec-derived tests seem to generalise better to unseen bugs.

Real upstream bugs, re-introduced

Each fix commit is reverted and every suite runs on the pre-fix code. Two bugs still escape today’s human tests:

402b351 mac_address accepted mixed ‘.’ and ‘-’ separators	Human tests	MutantHunter	Plain prompt (B1)
	✗ missed	✓ caught	✓ caught
41f3d6d hostname rejected single-character labels (RFC 1123 allows them)	Human tests	MutantHunter	Plain prompt (B1)
	✗ missed	✓ caught	✓ caught

Honest note: the plain prompt catches both too. Context only: on 8 url/email rows, human 7/8, MH 7/8, B1 5/8.

13 suspected violations, 5 confirmed bugs

13
suspected violations

9
confirmed entries

5
distinct bugs

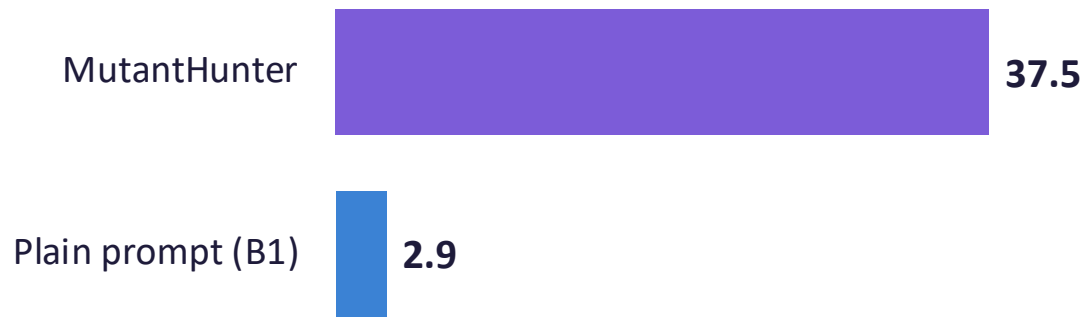
4 rejected as library design choices (single-character TLD, dotted MAC, port 0, strict_query)

Bug	Code (src/validators/)	Upstream
ISIN check digit is never verified	finance.py:34-51	known, #440
Hostname capped at 61 chars; RFC 1123 says 63	hostname.py:27-29	not reported
Email [IPv6:...] literal (RFC 5321) rejected	email.py:70	not reported
Quoted email local part with space or @ rejected	email.py:58, 93-95	not reported
URL with empty port rejected (RFC 3986 allows)	hostname.py:18-21	not reported

All 5 still present on upstream master (checked 2026-09-26); 4 are not reported upstream.

What it costs, and where it falls short

BOBCOINS FOR 6 MODULES



≈13×

the cost of the plain prompt

Median per module: 6.1 Bobcoins, 9.2 minutes

LIMITS WE REPORT

-  Budget: 6 of 12 planned modules run
-  Split B: no gain over the plain prompt
-  Exam C: a trend, not significant after correction
-  One library, one seed, simple operators

Every score comes with n , a 95% CI and its cost.

Try it: one sentence in IBM Bob



Evan

tool, skill, test generation



chu

evaluation, dashboard, video



Repository

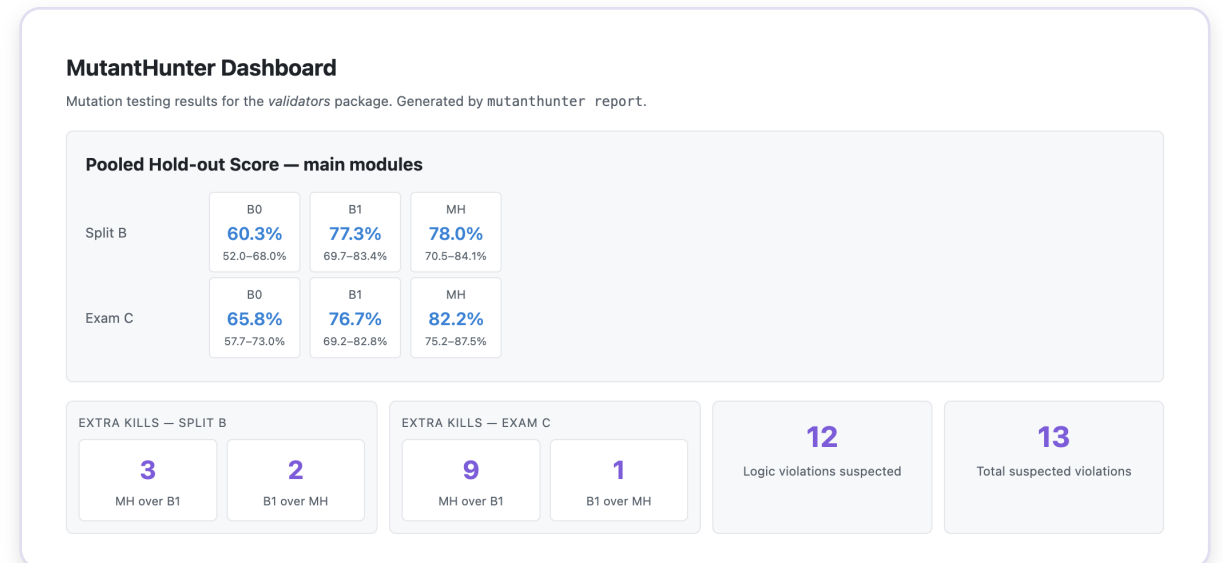
github.com/hanlinji4002/MutantHunter



Dashboard

hanlinji4002.github.io/MutantHunter

Spec-derived tests turn disagreements with the specification into reviewable bug reports.



The dashboard: pooled scores with 95% CIs, per-module results and every suspected violation.